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Development of a Simulation Model for Analysis and Evaluation of Human-Machine Collaboration in Assembly Cells

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1 – Introduction

Context and Motivation



Motivacional Background

Industrial assembly lines are increasingly relying on the distribution of tasks between robots and human operators. The challenge is to create a framework that supports setup decision-making and enhances efficiency and flexibility by enabling digital integration testing of collaborative robots in assembly lines.



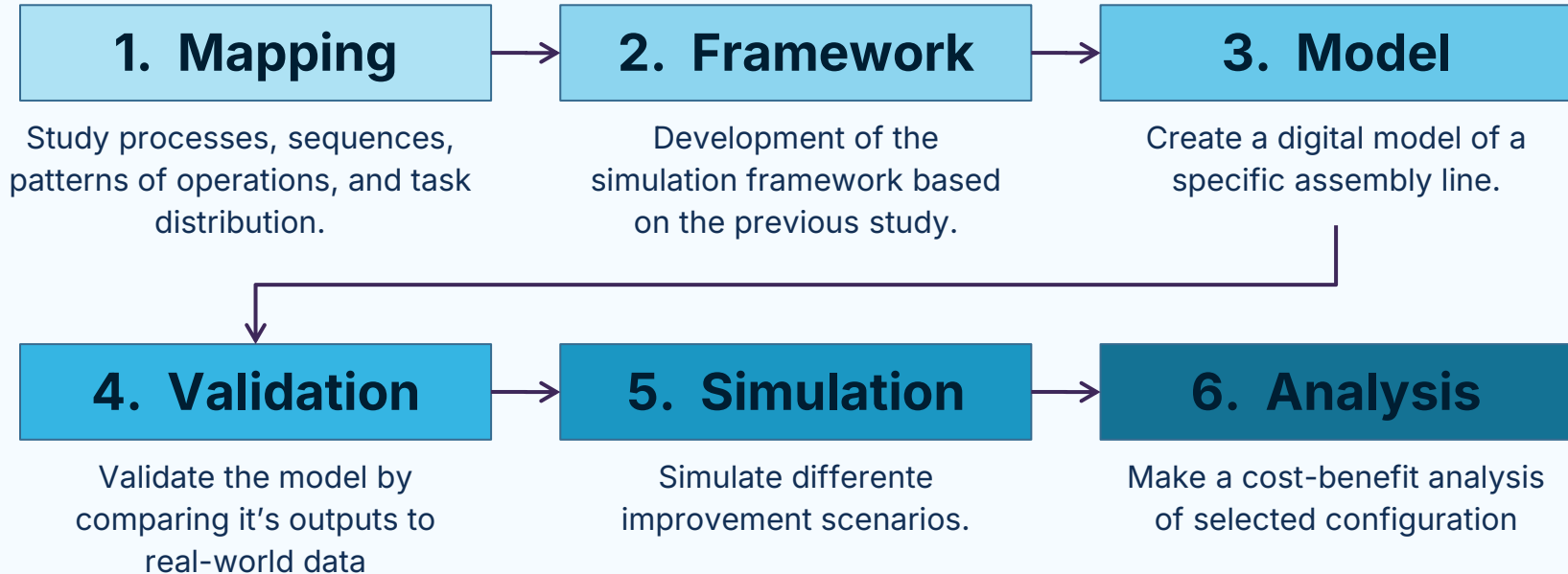
Development Environment

The dissertation project was developed within Bosch Security Systems, located in Ovar, which is composed of multiple assembly lines, with different degrees of automation.



1 – Introduction

Objectives





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1 – Introduction

Research Methodology



A comprehensive database is used to identify and collect relevant scientific literature.

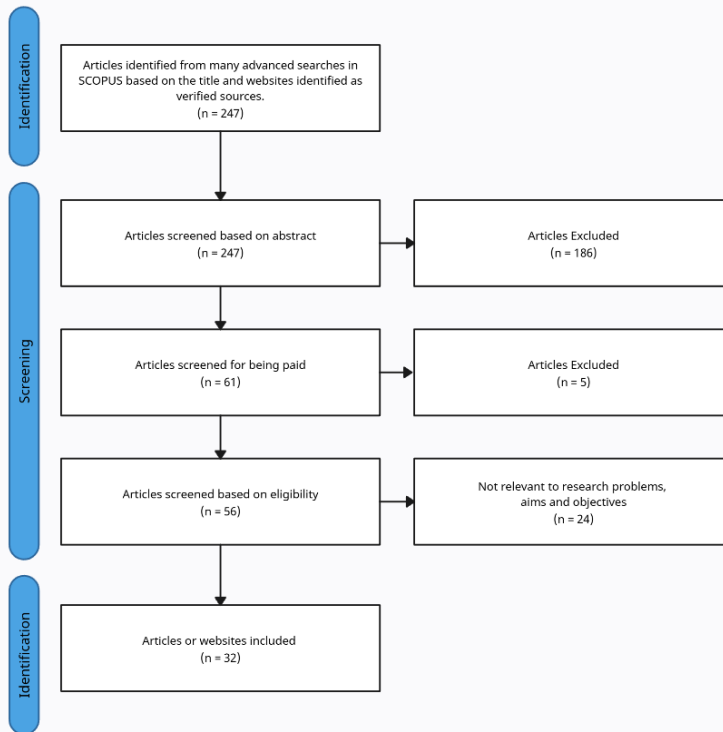


Reference manager to organize, annotate, and cite the selected articles efficiently.



AI-powered tool for summarizing, extracting insights, and querying uploaded articles.

PRISMA diagram - Literature review



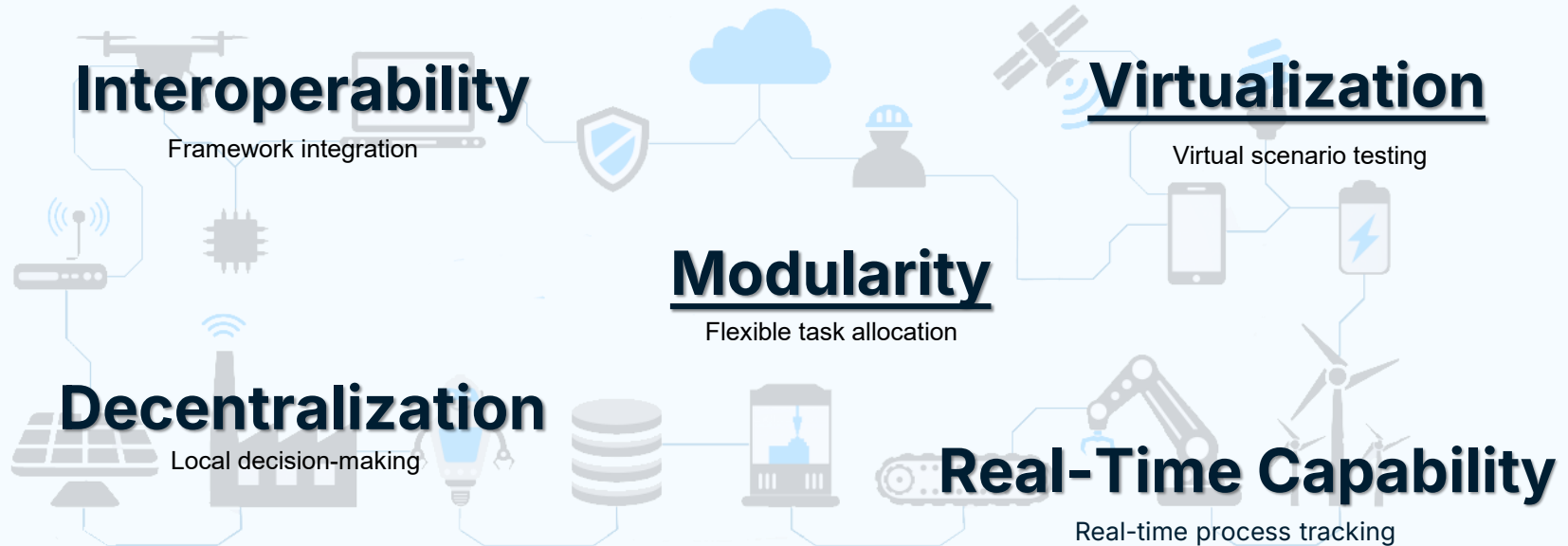


Search....

2 – *State of the Art*

Industry 4.0

J. Lettori et al. DOI: 10.3233/ATDE220671
Wojciech Lewicki et al. DOI: 10.3390/app14166957





2 – *State of the Art*

Human-Robot Collaboration



Why does it matter?

- Combines human flexibility + robot precision
- Improves efficiency and system adaptability
- Reduces heavy and repetitive payloads for operators

Collaboration Types

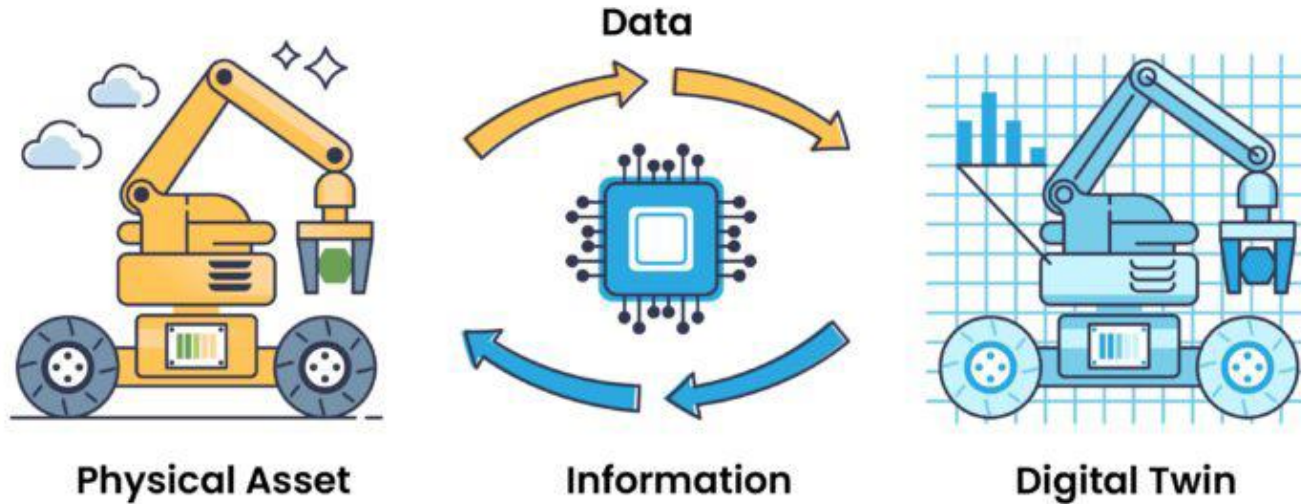
- **Independent:** Separate tasks
- **Sequential:** Same product, different tasks
- **Simultaneous:** Same product, same time
- **Supportive:** Human + robot assist



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2 – *State of the Art*

Digital Twins

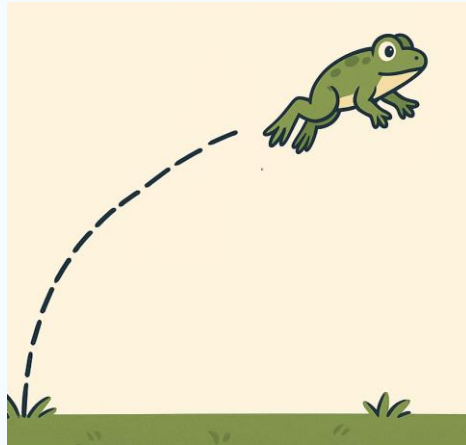




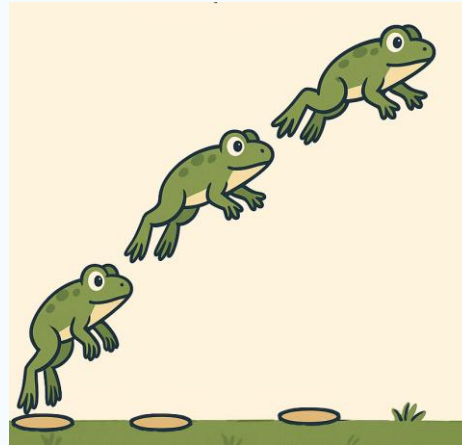
Search....

2 – *State of the Art*

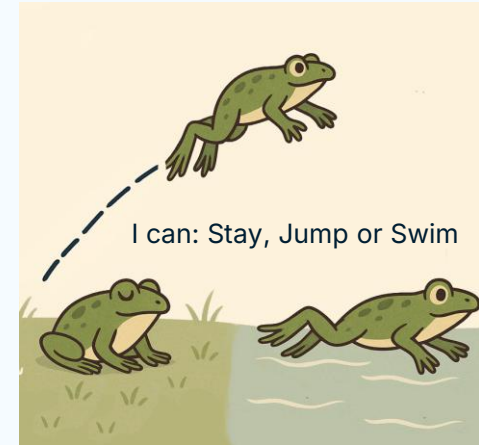
Process Modeling



Continuous



Discrete



Agent-Based



2 – *State of the Art*

Simulation Tools

Tool	License	Flexibility	Ease of Use	Visualization	Real-Time	Documentation
	Open	High	Moderate	Moderate	Moderate	Extensive
	Open	High	High	Limited	Limited	Extensive
	Paid	Moderate	High	Advanced	High	Comprehensive
	Paid	Moderate	High	Advanced	High	Comprehensive



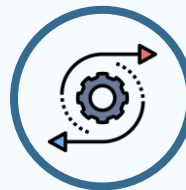
2 – *State of the Art*

Driven Decision Support



Definition

Combines data, models, and user interaction to enhance decision-making.



Hybrid Production

Supports human-robot collaboration, resource allocation, and bottleneck identification.



Purpose

Enables “what-if” analyses of production scenarios without affecting real operations



Benefits

Enhances scenario evaluation and supports iterative learning to strengthen operational planning.



3 – *Development*

Design Principles and Tool Justification

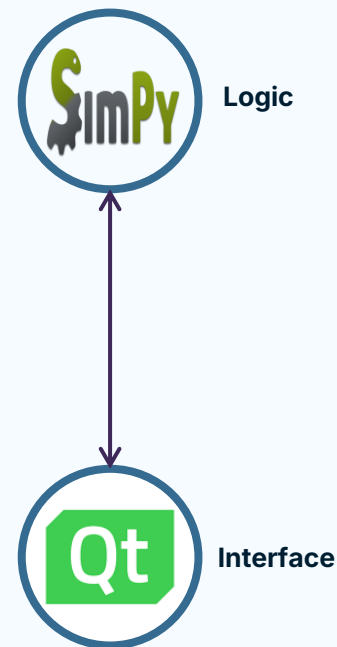
What are the goals?

The primary objective of the proposed framework is to provide a flexible, configurable, and realistic simulation environment to support scenario analysis and decision-making.

Why SimPy and Qt ?

SimPy was chosen because it is an open-source Python library that offers a flexible and precise engine for modeling processes.

Qt complements this by providing tools for creating an intuitive graphical interface, which makes the framework interactive and accessible.

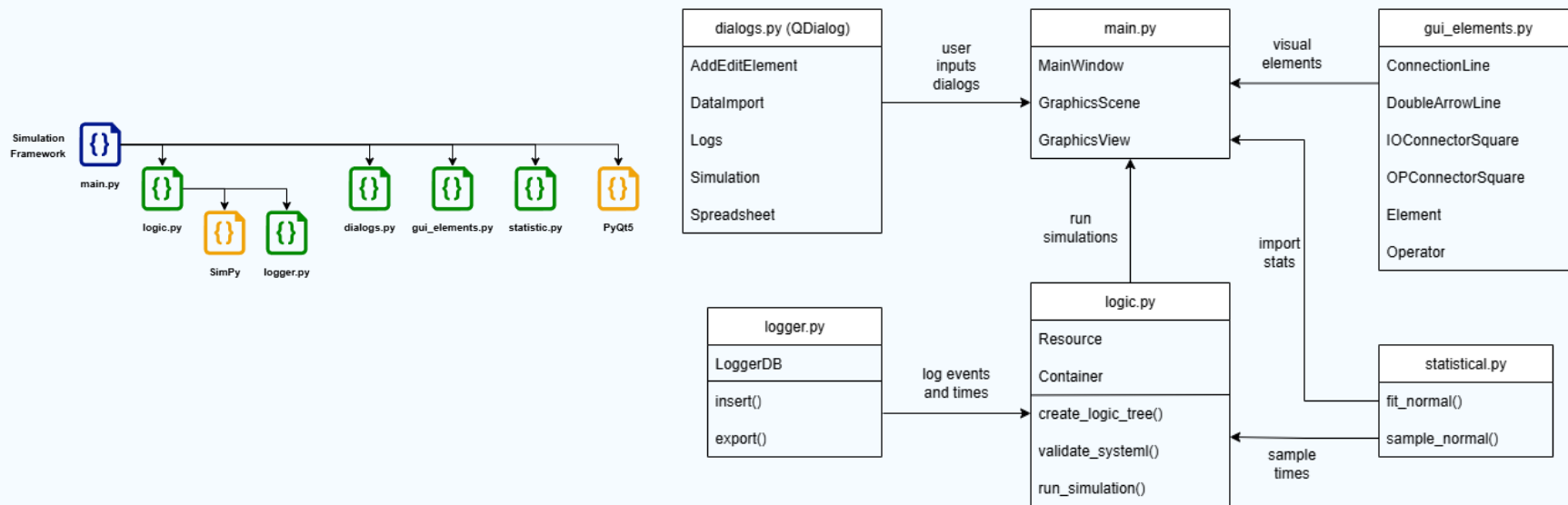




Search....

3 – Development

Architecture





Search....

3 – Development

Logic with SimPy – Process-Based Modeling

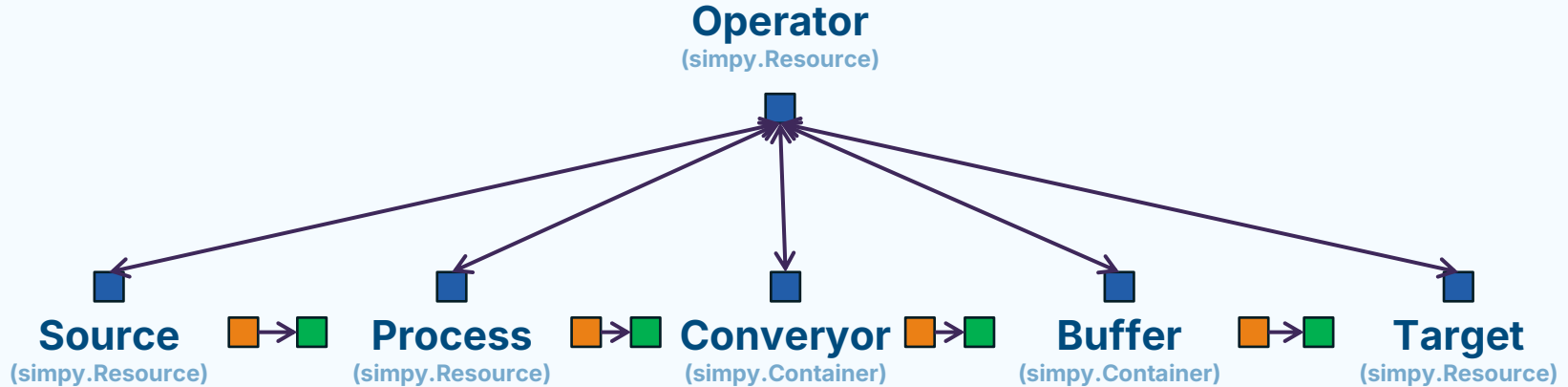
To be done



Search....

3 – Development

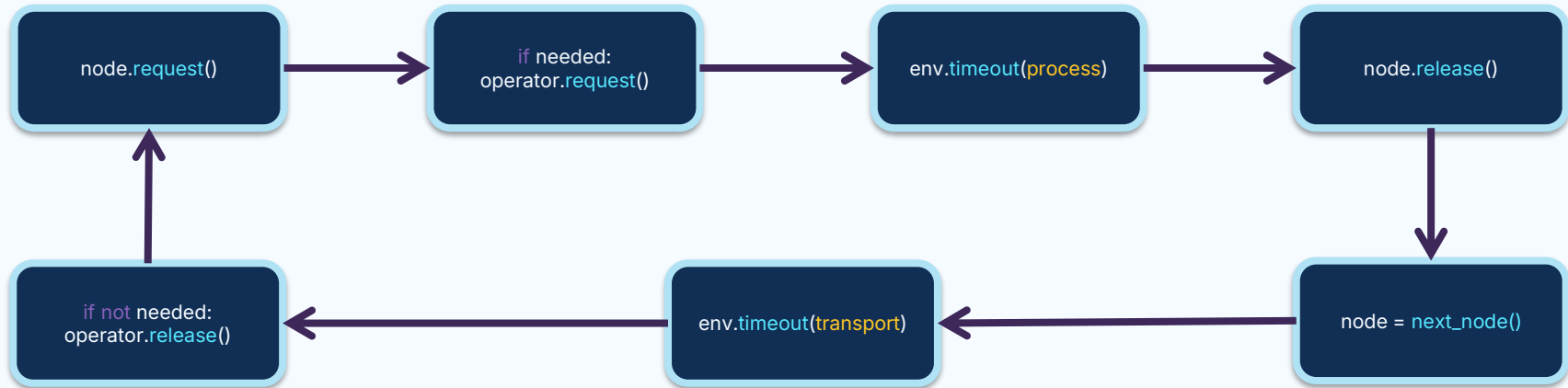
Logic with SimPy – Simulation Entities





3 – Development

Logic with SimPy – Simulation Flow





3 – Development

Logic with SimPy – Errors and Failures Simulation



Piece Conformity Errors

- Each process has a configurable defect rate
- Detected defective parts are immediately rejected



Simulation Behavior

- Non-conformities occur probabilistically
- Failures and maintenance trigger at the respective threshold.
- Preventive Maintenance



Failures & Maintenance

- **MTBF** – avg. time between failures
- **MTTR** – avg. repair time
- **MTBM** – avg. time between maintenance
- **TSLM** – time since last maintenance
- **MTM** – avg. maintenance duration



Event Logging

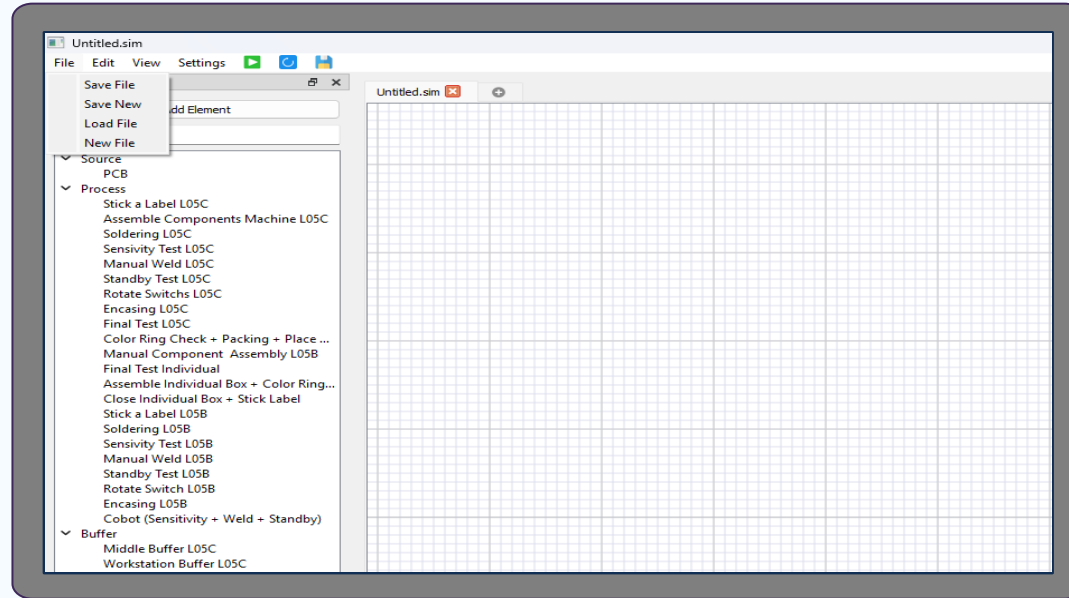
- Every event from piece movement and operator interaction is logged



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3 – Development

GUI with Qt – Main Window



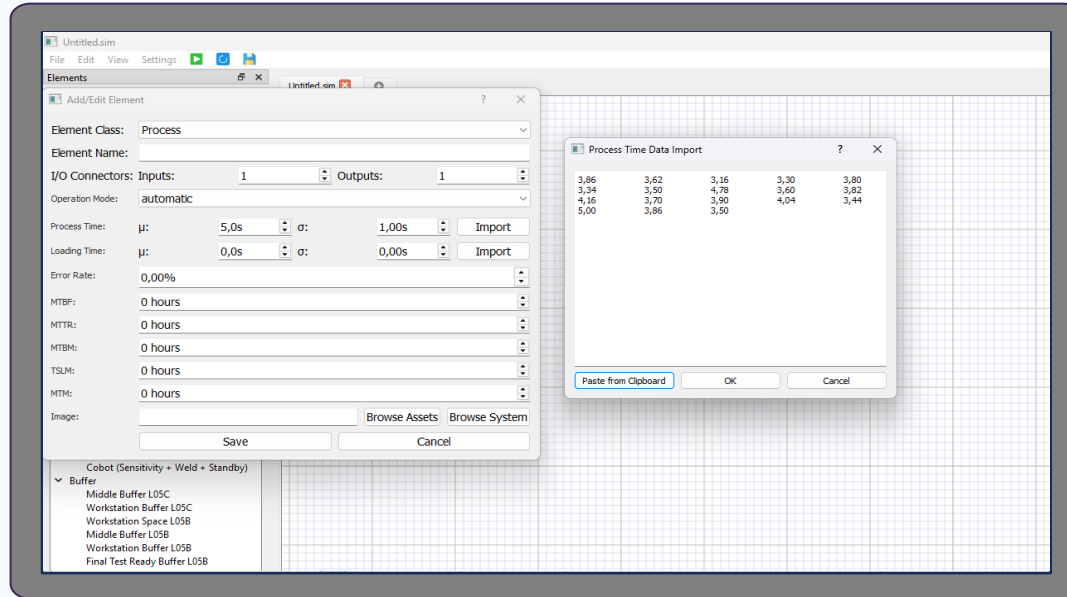
Simulation Framework Main Window



Search....

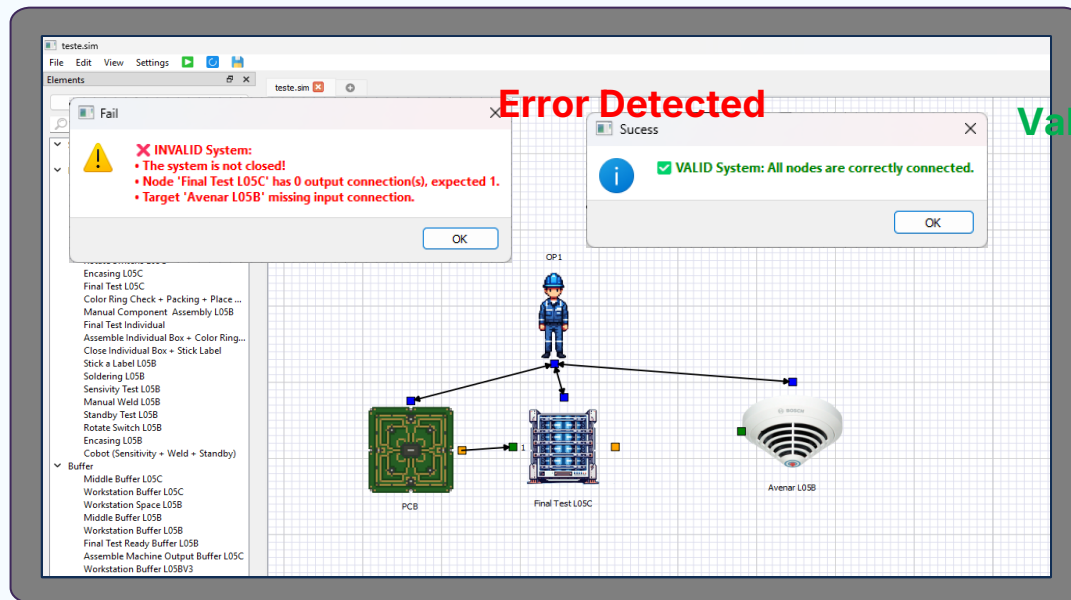
3 – Development

GUI with Qt – Adding Elements



Add/Edit Element with Import Data Window

GUI with Qt – Model Validation



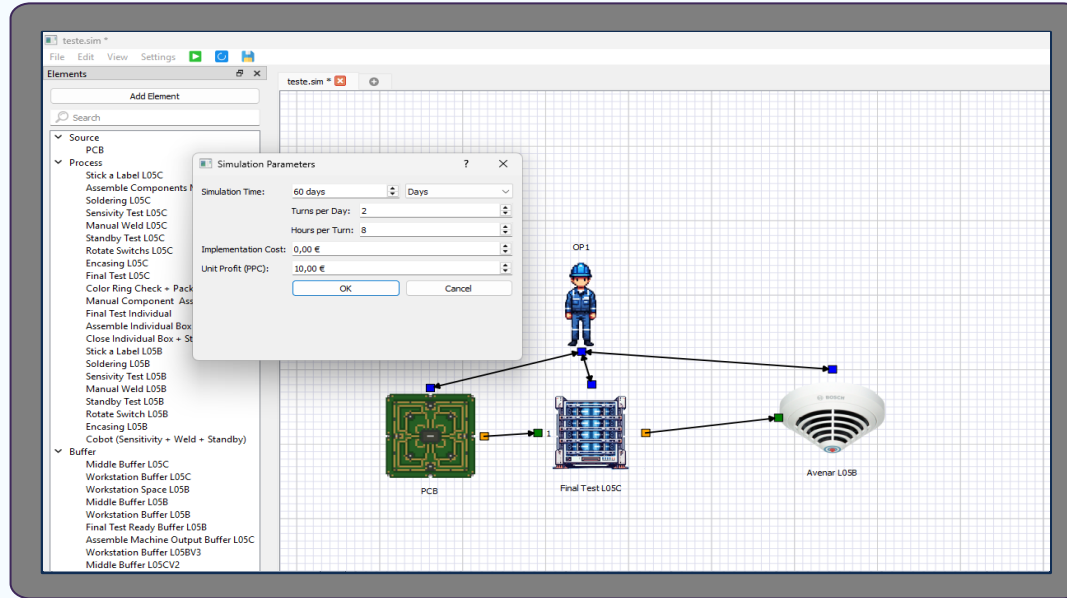
Validation Return Options



Search....

3 – Development

GUI with Qt – Simulation Parameters



Simulation Parameters



3 – Development

Output Metrics - Flows

	ID ↑	NODE	CYCLE_TIME	PROCESS_TIME	COMMENT
1	1.1.1.1	PCB	0.0	0.0	In
2	1.1.1.1	PCB	0.0	0.0	Transport to Stick a Label
3	1.1.1.1	Stick a Label	0.0	0.0	Input Inserted Need 1 Have:1
4	1.1.1.1	PCB	0.0	0.0	Out
5	1.1.1.1	Stick a Label	0.0	0.0	In
6	1.1.1.1	Stick a Label	0.0	0.0	Manual Work Start
7	1.1.1.1	Stick a Label	3.3	3.3	Manual Work End
8	1.1.1.1	Stick a Label	3.3	3.3	Transport to Assemble Machine Input Conveyor
9	1.1.1.1	Stick a Label	3.3	3.3	Out
10	1.1.1.1	Assemble Machine Input Conveyor	3.3	3.3	In
11	1.1.1.1	Assemble Machine Input Conveyor	3.3	3.3	Conveyor Transport
12	1.1.1.1	Assemble Machine Input Conveyor	3.3	3.3	Transport to Assemble Components
13	1.1.1.1	Assemble Components	3.7	3.7	Input Inserted Need 1 Have:1
14	1.1.1.1	Assemble Machine Input Conveyor	3.7	3.7	Out

Simulation Piece Flow Spreadsheet

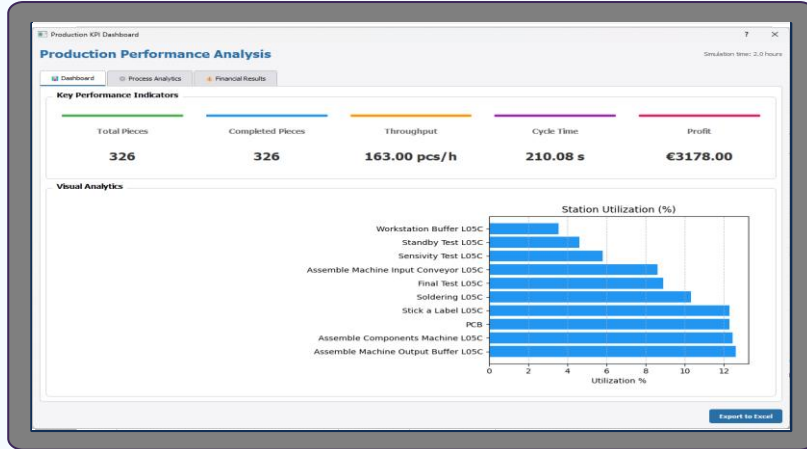
	ID	NAME	NODE	TYPE	PROCESS_TIME ↑	PIECE
1	26.0	OP1	PCB	Request	0.0	1.1.1.1
2	26.0	OP1	Stick a Label	Release	0.0	1.1.1.1
3	26.0	OP1	Stick a Label	Request	0.0	1.1.1.1
4	26.0	OP1	Stick a Label	Hold for Conveyor	3.3	1.1.1.1
5	26.0	OP1	Stick a Label	Release	3.3	1.1.1.1
6	26.0	OP1	Assemble Machine Input Conveyor	Request	3.3	1.1.1.1
7	26.0	OP1	Assemble Components	Release	3.7	1.1.1.1
8	26.0	OP1	PCB	Request	3.7	1.1.2.1
9	26.0	OP1	Stick a Label	Release	3.7	1.1.2.1
10	26.0	OP1	Stick a Label	Request	3.7	1.1.2.1
11	26.0	OP1	Stick a Label	Hold for Conveyor	7.5	1.1.2.1
12	26.0	OP1	Stick a Label	Release	7.5	1.1.2.1
13	26.0	OP1	PCB	Request	7.5	1.1.3.1
14	26.0	OP1	Stick a Label	Release	7.5	1.1.3.1

Operator Workflow Spreadsheet



3 – Development

Output Metrics – KPIs and Efficacy



Production KPIs Dashboard

Process Efficacy Analysis

Node	Avg Manual	Avg Automatic	Avg Loading	% of Total Time
1 Stick a Label ...	3.67	0.00	3.94	6.9%
2 Assemble ...	0.00	19.70	0.00	17.5%
3 Assemble ...	0.00	0.00	4.15	3.7%
4 Soldering L05C	0.00	13.83	4.78	16.3%
5 Middle Buffer ...	0.00	0.00	2.56	2.2%
6 Sensivity Test ...	0.00	10.45	0.00	9.1%
7 Manual Weld ...	4.24	0.00	1.55	5.1%
8 Standby Test ...	0.00	8.25	0.00	7.2%
9 Rotate Switchs ...	2.35	0.00	0.00	2.1%
10 Encasing L05C	3.44	0.00	0.00	3.0%
11 Workstation ...	0.00	0.00	2.25	2.0%
12 Final Test L05C	0.00	15.96	0.00	13.9%
13 Color Ring Che...	12.69	0.00	0.00	11.0%

Export to Excel

Process Efficacy Spreadsheet



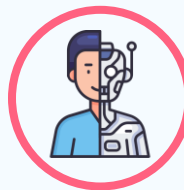
3–*Development*

Framework Limitations



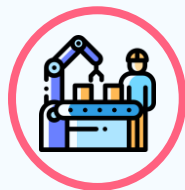
Operator Behavior

- Real operators level of experience and skill level are not modeled.
- Operator simulation is linear and deterministic.



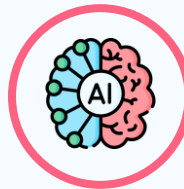
Simulation of CoBots

- Cobot behavior was modeled based on field observations and internal assumptions.



Physical World Modeling

- The framework lacks physical aspects such as safety procedures or ergonomic factors.



Digital Twins and AI

- The framework does not include automatic balancing, intelligent routing, or optimization features.



4 – Case Study

Context

Industrial Context

Bosch Security Systems, S.A., located in Ovar, operates a significant manufacturing facility that produces security and fire detection systems. The shop floor features multiple assembly lines dedicated to assembling complex products.

Lines under Study

L05B – Produces the Avenar fire detector in a three-operator regime, which relies almost entirely on human work.

L05C – Is a more automated version of L05B

Why these two lines?

Same product, two different profiles → clear comparison study for manual vs automated HRC lines.



Avenar Fire Detector



4 – Case Study

Modeling Methods

i) Conceptual Models

- Process flow diagrams
- Built from on-site observations and engineer feedback
- Excel timing mapped to tasks

ii) Digital Models

- Modular discrete-event blocks per station
- Normal distributions for time simulation
- Blocks connect following the real flow

Operator Modeling

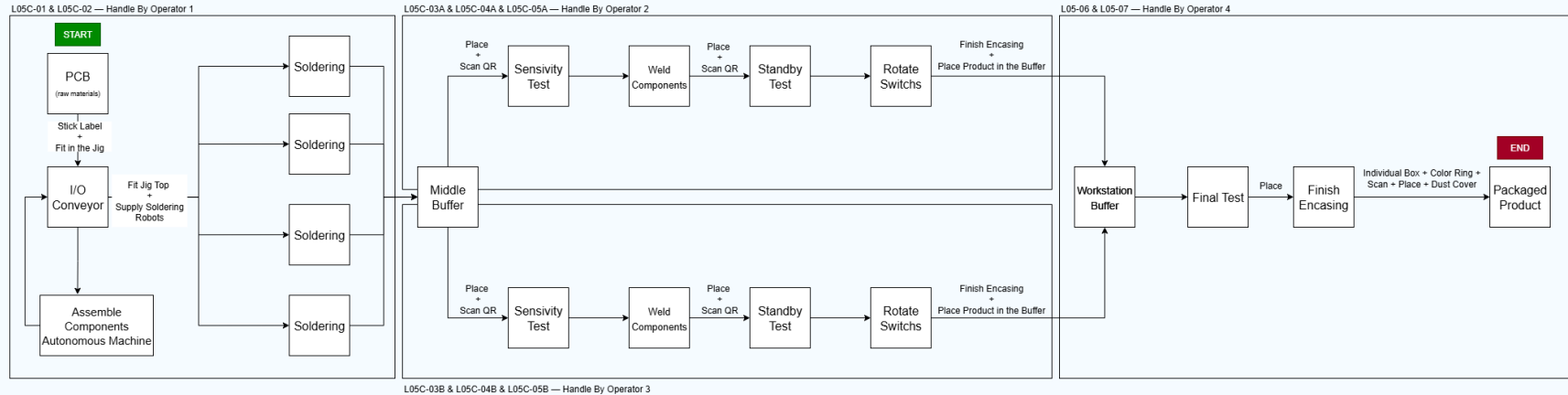
- Operator as a resource with availability and cost
- Task embedded in process blocks
- Each operator task is a linked process



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4 – Case Study

Assembly Line L05C - Diagram



Block Diagram of L05C



4 – Case Study

Assembly Line L05C – Processing Times

Texto de suporte

#	Process	Average Time (s)
1	Stick the label	3.73
2	Fit the PCB in the jig	4.04
3	Assemble Components	19.70
4	Fit Jig Top + Supply Welding Robot	4.23
5	WeldComponents	13.875
6	RemoveJig + Place in the Buffer	4.81
7	Place in the Sensitivity Test + Scan Barcode	2.64
8	Sensitivity Test	10.54
9	Place in the Welder + Weld Components	4.305
10	Place in the Standby Test + Scan Barcode	1.57
11	Execute Standby Test	8.335
12	Rotate Switches	2.19
13	Finish Encasing + Place in Buffer	3.505
14	Stick Color Ring + Place in the Final Test	2.27
15	Execute Final Test	16
16	Individual Box + Color Ring + Scan + Dust Cover + Place in Collective	11.97

L05C Bosch Provided Data

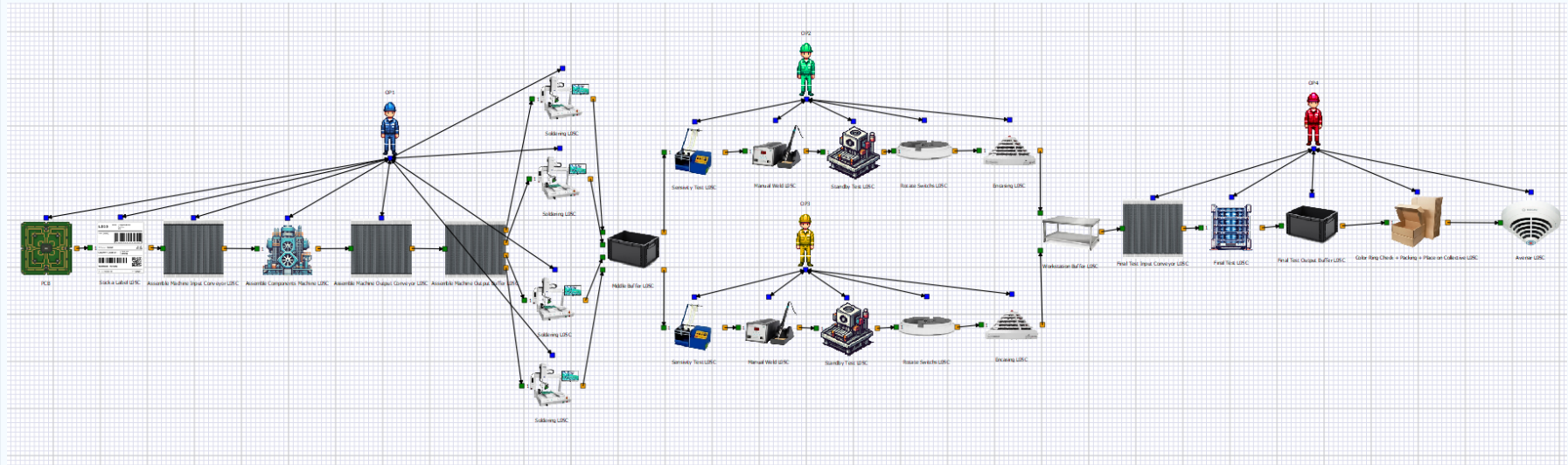


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* Wrong image in the dissertation document

4 – Case Study

Assembly Line L05C – Simulation Model



Simulation Model of L05C

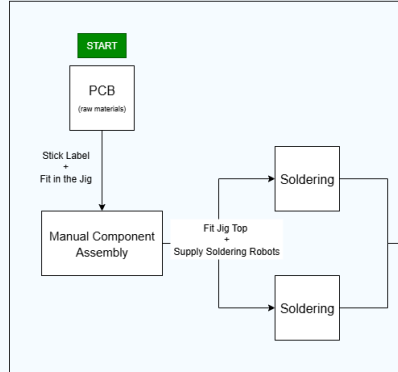


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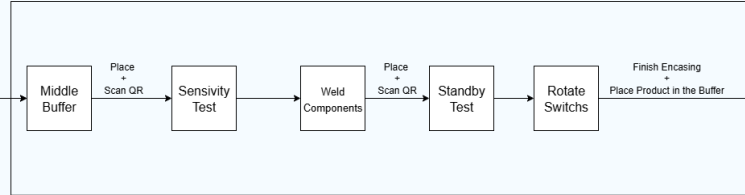
4 – Case Study

Assembly Line L05B - Diagram

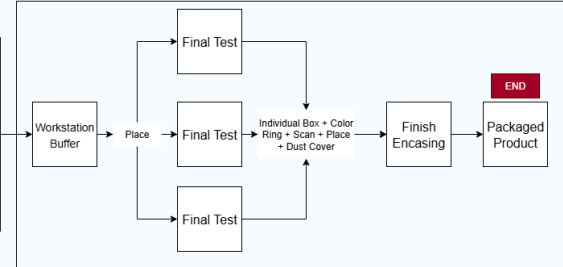
L05B-01 & L05B-02 — Handle By Operator 1



L05B-03 & L05B-04 & L05B-05 — Handle By Operator 2



L05B-06 & L05B-07 — Handle By Operator 3



Block Diagram of L05B



4 – Case Study

Assembly Line L05B – Processing Times

Texto de suporte

#	Process	Average Time (s)
1	Stick the label	3.73
2	Fit the PCB in the jig	4.36
3	Manual Component Assembly	24.70
4	Fit Jig Top + Supply Welding Robot	4.23
5	WeldComponents	13.88
6	RemoveJig + Place in the Buffer	5.54
7	Place in the Sensitivity Test + Scan Barcode	2.45
8	Sensitivity Test	10.54
9	Place in the Welder + Weld Components	4.56
10	Place in the Standby Test + Scan Barcode	1.72
11	Execute Standby Test	8.42
12	Rotate Switches	2.14
13	Finish Encasing + Place in Buffer	3.505
14	Place product in Final Test	2.56
15	Execute Final Test	24
16	Assemble Individual Box + Color Ring Check + Scan + Place	11.97
17	Close Individual Box + Stick Label + Place	5.96

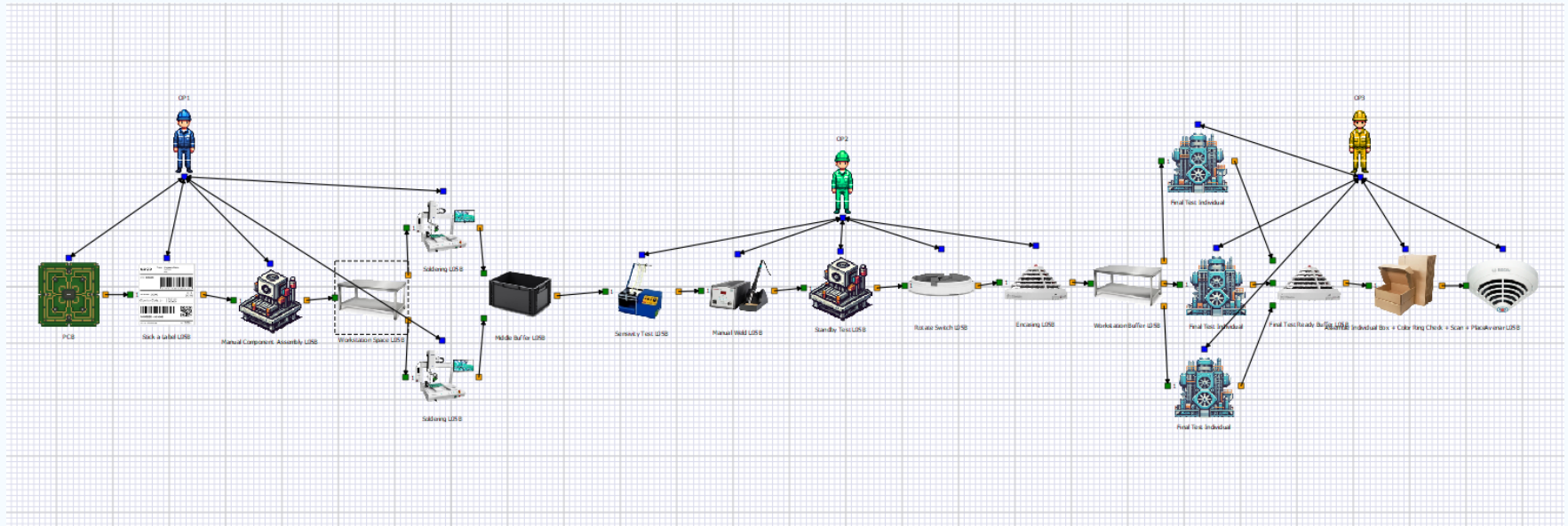
L05B Bosch Provided Data



Search....

4 – Case Study

Assembly Line L05B – Simulation Model



Simulation Model of L05B



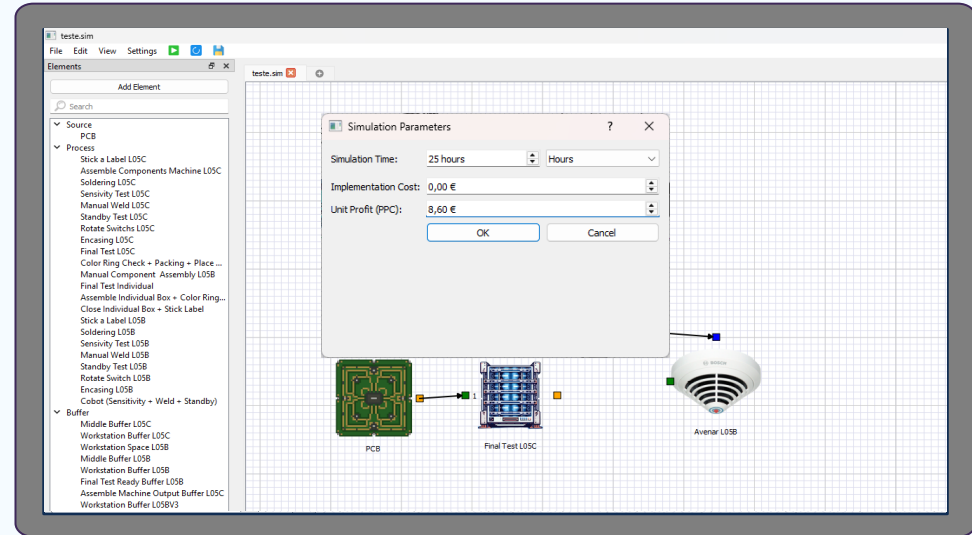
4 – Case Study

Simulation Setup

Each scenario was executed for 25 simulated hours to match Bosch's minimum baseline window for KPI extraction.

The economics were parameterized using PPC (Bosch's unit profit per assembled unit) to translate throughput into revenue and estimate ROI and payback across scenarios.

This setup ensures that technical KPIs are directly comparable with other scenarios.

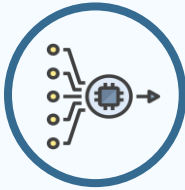


Simulation Setup Window



5 – Results

Evaluation Metrics



Throughput

Total number of average finished products per hour in a given simulation.



Overall Cycle Time (CT_{med})

Represents the average time interval between the completion of two consecutive pieces.



Processing Time

Total time each piece spends being processed, in each process.



Value-Added Time

The execution of added value operations in each given node.



5 – Results

L05C Case Study Results – Utilization per Station

Texto de suporte

Station	Time (s)	% Utilization
PCB	22.73	7.75%
Stick a Label L05C	22.72	7.75%
Assemble Components Machine L05C	22.67	7.73%
Assemble Machine Input Conveyor L05C	18.56	6.33%
Assemble Machine Output Buffer L05C	109.77	37.43%
Assemble Machine Output Conveyor L05C	21.24	7.24%
Soldering L05C	18.64	6.36%
Middle Buffer L05C	2.57	0.87%
Sensitivity Test L05C	10.52	3.59%
Manual Weld L05C	5.80	1.98%
Standby Test L05C	8.34	2.84%
Rotate Switches L05C	2.35	0.80%
Encasing L05C	3.44	1.17%
Workstation Buffer L05C	6.44	2.19%
Final Test L05C	16.11	5.49%
Final Test Input Conveyor L05C	1.37	0.47%

Utilization per Station L05C



5 – Results

L05C Case Study Results – Added Time Value

Station	Manual (s)	Automatic (s)	Loading (s)	% of Total Time
Stick a Label L05C	3.65	0.00	3.96	6.7%
Assemble Components Machine L05C	0.00	19.70	0.00	17.3%
Assemble Machine Output Buffer L05C	0.00	0.00	4.16	3.7%
Soldering L05C	0.00	13.84	4.76	16.3%
Middle Buffer L05C	0.00	0.00	2.54	2.2%
Sensitivity Test L05C	0.00	10.44	0.00	9.2%
Manual Weld L05C	4.24	0.00	1.55	5.1%
Standby Test L05C	0.00	8.25	0.00	7.2%
Rotate Switchs L05C	2.34	0.00	0.00	2.1%
Encasing L05C	3.45	0.00	0.00	3.0%
Workstation Buffer L05C	0.00	0.00	2.24	2.0%
Final Test L05C	0.00	15.95	0.00	14.0%
Color Ring Check + Packing + Place on Collective L05C	12.66	0.00	0.00	11.1%

Added Time Value per Station on L05C



5 – Results

L05C Case Study Results – Comparison with Real Data

Side-by-side comparison between simulated L05C results and Bosch real data across three 25-hour shifts, using Throughput and CT_{med} as validation KPIs.

Metric	Shift 1	Shift 2	Shift 3	Average	Simulation
Throughput (pcs/hr)	146.44	152.51	152.11	150.30	158.48
CT_{med} (s)	24.61	23.62	23.68	23.97	22.72

Simulation aligns closely with reality, with a deviation of ~5.4%. The gap is expected since human behavior isn't modeled, yet the model captures L05C's dynamics and is valid for scenario testing and optimization



5 – Results

L05B Case Study Results – Utilization per Station

Texto de suporte

Station	Avg Time (s)	% Utilization
PCB	34.79	1.16%
Stick a Label L05B	13.12	0.44%
Manual Component Assembly L05B	30.70	1.02%
Workstation Space L05B	68.04	1.02%
Soldering L05B	31.56	1.05%
Middle Buffer L05B	1970.46	65.70%
Sensivity Test L05B	27.15	0.91%
Manual Weld L05B	31.71	1.06%
StandbyTest L05B	29.21	0.97%
Rotate Switch L05B	31.46	1.05%
Encasing L05B	31.90	1.06%
Workstation Buffer L05B	678.34	22.62%
Final Test Individual L05B	20.95	0.70%

Utilization per Station L05B



5 – Results

L05B Case Study Results – Added Time Value

Station	Manual (s)	Automatic (s)	Loading (s)	% of Total Time
Stick a Label L05B	0.00	3.76	3.95	6.60%
Manual Component Assembly L05B	21.59	0.00	0.00	18.30%
Workstation Space L05B	0.00	0.00	3.15	2.70%
Soldering L05B	0.00	13.80	5.45	16.30%
Middle Buffer L05B	0.00	0.00	2.44	2.00%
Sensitivity Test L05B	0.00	10.46	0.00	8.50%
Manual Weld L05B	3.76	0.00	1.55	4.30%
Standby Test L05B	0.00	8.25	0.00	6.70%
Rotate Switch L05B	2.14	0.00	0.00	1.70%
Encasing L05B	0.00	3.95	0.00	3.20%
Workstation Buffer L05B	0.00	0.00	2.74	2.20%
Final Test Individual	0.00	20.95	0.00	16.90%
Assemble Individual Box + Color Ring Check + Scan + Place	7.05	0.00	5.85	10.40%

Added Time Value per Station on L05B



5 – Results

L05B Case Study Results – Comparison with Real Data

Side-by-side comparison between simulated L05B results and Bosch real data across three 25-hour shifts, using Throughput and CT_{med} as validation KPIs.

Metric	Shift 1	Shift 2	Shift 3	Average	Simulation
Throughput (pcs/hr)	92.94	96.14	95.76	94.95	103.56
CT_{med} (s)	35.32	34.77	35.37	35.15	34.76

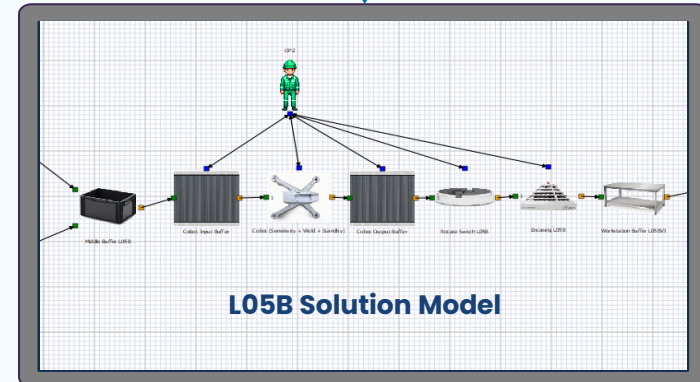
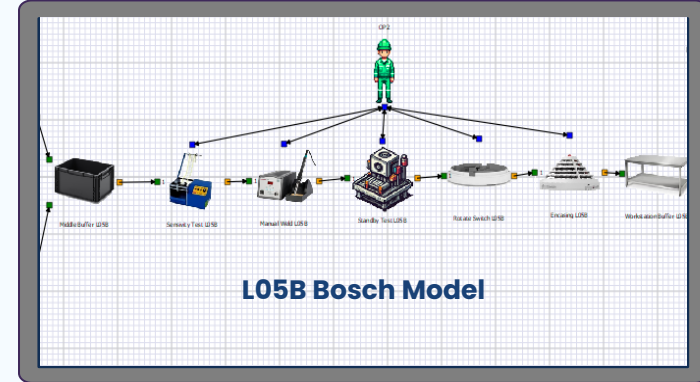
The simulation aligns closely with reality, with a deviation of only 9%. Because individual human behaviors are not modeled, and the flow assumes uninterrupted operation, slight output overestimation is expected.



5 – Results

Solution for L05B – Integration Description

- Consolidates **Sensitivity Test + Resistor Welding + Stanby Test** into one coordinated loop.
- A **cobot** manages the testing, loads components into an autonomous welder, transfers them to the standby test, and outputs to a slide buffer.
- Operates within an isolated acrylic or glass enclosure, with the operator feeding into an input buffer and receiving from the slide buffer.
- Loading time of N(2.43 s, 0.41 s).
- Processing time of N(17.42 s, 0.80 s).





** Rough Estimates Based on Feedback from
Bosch Industrial Engineers*

5 – Results

Solution for L05B – Implementation Cost

Cost Component	Estimated Cost (€)
Cobot arm (including controller base software)	30,000 – 50,000
Custom gripper / end effector	5,000 – 10,000
Safety enclosure and physical isolation	5,000 – 15,000
Integration, programming, and validation	15,000 – 30,000
Layout adaptation and auxiliary equipment	5,000 – 10,000
Operator training and knowledge transfer	3,000 – 7,000
Total Estimated Cost	63,000 – 122,000

Estimating that the integration would require 5 working days, and assuming an effective daily workload of 7.70 hours.

Given an average production of 94.95 pieces an hour and a PPC of 8.6€, the temporary suspension would cost about **31,438€**.

The final implementation would cost at a minimum of:

94,438 €



5 – Results

Solution for L05B – Implementation Results

* % improvement value wrong in the
dissertation document

Side-by-side comparison between the real, the simulated and the solution L05B throughput
and CT_{med} as validation KPIs.

L05B	Real	Simulation	Solution
Throughput (pcs/hr)	94.95	103.56	123.40
CT_{med} (s)	35.15	34.76	29.17

The solution represents a 19% improvement in production throughput.



5 – Results

Solution for L05B – Process Time

Texto de suporte

Station	Avg Time (s)	Utilization
PCB	29.25	1.16%
Stick a Label L05B	21.59	0.38%
Manual Component Assembly L05B	21.59	1.08%
Workstation Space L05B	3.30	0.16%
Soldering L05B	15.82	0.79%
Middle Buffer L05B	810.31	40.42%
Cobot Input Buffer L05BV2	29.19	1.46%
Cobot (Sensitibity + Weld + Standby) L05BV2	96.31	4.80%
Cobot Output Buffer L05BV2	268.51	13.40%
Rotate Switch L05B	29.09	1.45%
Encasing L05B	29.80	1.49%
Workstation Buffer L05B	642.70	32.06%
Final Test Individual L05B	20.95	1.05%

Utilization per Station on L05B Solution



5 – Results

Solution for L05B – Added Time Value

Station	Manual (s)	Automatic (s)	Loading (s)	% of Total Time
Stick a Label L05B	0.00	3.74	3.97	8.30%
Manual Component Assembly L05B	21.59	0.00	0.00	23.10%
Workstation Space L05B	0.00	0.00	3.15	3.40%
Soldering L05B	0.00	13.80	1.95	16.80%
Middle Buffer L05B	0.00	0.00	0.5	0.5%
Cobot (Sensitivity + Weld + Standby)	0.00	12.33	0.00	11.30%
Rotate Switch L05B	2.11	0.00	0.00	1.90%
Encasing L05B	0.00	3.92	0.00	3.50%
Workstation Buffer L05B	0.00	0.00	2.76	2.40%
Final Test Individual	0.00	20.95	0.00	17.90%
Assemble Individual Box + Color Ring Check + Scan + Place	7.07	0.00	5.83	11.00%

Added Time Value per Station on L05B Solution



5 – Results

Cost-Benefit Considerations

Texto de suporte

Metric	Value
Baseline Daily Output	797.41 pcs
Improved Daily Output	950.18 pcs
Baseline Days to Produce 2589 pcs	3.25 days
Improved Days to Produce 2589 pcs	2.72 days
Time Saved	0.52 days
Extra Pieces in Equal Time	484 pcs
Extra Profit Generated	4,163.53 €
Implementation Cost	94,438 €
Estimated Payback Time	73.26 days



6 – Conclusions

Critical Assessment



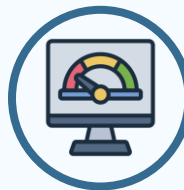
Configurational Agility

Core abstractions let us reconfigure assembly lines fast, like adding a cobot and reassigning OP2 needed minimal changes.



Fidelity vs Practicality

The more accurate and realistic the data, the higher the fidelity of the simulation output will be.



Computational Efficiency

The DES kernel (Simpy) handles queuing, resource contention, and task preemption, however, high-load simulations have not been thoroughly tested.



Human Factors Exclusion

Ergonomics and fatigue are not considered leading to potential operator-centric design gaps.



6 – *Conclusions*

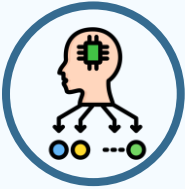
Alignment with Dissertation Objectives

1. Map the main processes and operational routing in the Bosch Ovar assembly cells that could be automated using robotic systems.
2. Develop a simulation model using an appropriate platform (e.g., Arena, AnyLogic, FlexSim).
3. Simulate various operational scenarios and task-distribution strategies between operators and robots, taking into account production times and cadence, operating costs, and assembly cell flexibility.
4. Perform scenario analyses of operator–robot task allocation, including a sensitivity analysis based on different levels of operator experience and training.
5. Conduct a cost–benefit analysis of the proposed solution.
6. Propose improvements, optimizations, and best practices for the current production processes in Bosch assembly cells, based on the simulation results.



6 – *Conclusions*

Future Work



Broadened Resource Modeling

The framework can be expanded to support a wider range of industrial elements by incorporating machines, dynamic buffers, and more reliable cobots.



Use-Case Expansion

Strengthening the framework's credibility requires broader deployment and testing.



Results Automation

Future development should also prioritize the automation of result presentation dynamics to enhance the extraction of KPIs.



Pathway to Digital Twin

Continuous development for real-time data integration and real-time visualization.



Search....

Thanks for the Attention

Development of a Simulation Model for Analysis and Evaluation of Human-Machine Collaboration in Assembly Cells

